

Percolation path and dielectric-breakdown-induced-epitaxy evolution during ultrathin gate dielectric breakdown transient

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A physical model has been developed which complies with the experimental observation on the failure mechanism of ultrathin gate oxide breakdown during constant voltage stress. Dynamic equilibrium needs to be established between the percolation conductive path and the dielectric breakdown induced epitaxy (DBIE) formation during gate dielectric breakdown transient. The model is capable of linking the percolation model, soft breakdown, and hard breakdown to the DBIE growth for a variety of stress conditions and gate oxide thickness without involving new empirical parameters. © 2003 American Institute of Physics. [DOI: 10.1063/1.1611649]

It has been shown that during gate oxide breakdown transient, various microstructural damages and changes can occur within the transistor structures.^{1–3} Among the physical changes observed experimentally, dielectric breakdown induced epitaxy (DBIE) is believed to be the failure mechanism that is of critical importance in understanding the gate oxide breakdown. Under constant voltage stress (CVS) mode, isolated narrow transistors show impromptu and irreversible microstructural changes in both the cases of soft breakdown (SBD) and hard breakdown (HBD). Contradicting to the conventional belief which considers the defects and damages occur only within gate dielectric materials during breakdown, the physical and microstructural changes to the gate electrode (polysilicon) and substrate channel (single crystal Si) play an important role in a breakdown event.

It is shown that a comprehensive understanding can be established, based on the percolation model and the experimental evidences obtained from our physical analysis, to explain the occurrence of DBIE during SBD and HBD. The dynamic and delicate equilibrium between the percolation path evolution and DBIE growth during breakdown transient is the key in understanding the ultrathin gate dielectric breakdown mechanism.

Several types of transistors with gate oxide thickness ranging from 33 to 16 Å fabricated using standard 0.18/0.15/0.13 μm salicide complementary metal–oxide–semiconductor processes were used for this study. Narrow transistors of $0.18 \times 0.4 \mu\text{m}^2$ and $0.15 \times 0.3 \mu\text{m}^2$ were stressed using constant voltage. Various current compliance

settings ranging from 100 nA to 1 mA were employed. The physical analysis of the devices after the *first* breakdown stressing was carried out using high-resolution transmission electron microscopy (TEM).¹ The measurements of the BD transients were taken by using a configuration with a current amplifier with a 1 MHz bandwidth as discussed in Ref. 4.

Numerous physical damages observed after gate oxide breakdown have been reported previously.^{1–3} Two failure mechanisms are noted: dielectric breakdown induced metal migration and DBIE. When transistors are stressed with low levels of current compliance (such as 10 μA or less), DBIE is found to be the dominant failure mechanism during SBD, while most of the other physical damages diminish.^{2–3} In addition, DBIE was found to be stress polarity dependent, and electrothermal migration could be the driving force for its formation.² Regardless of the stress mode, as the SBD DBIE hillock from the cathode extruding into the gate oxide and subsequently into the other electrode (i.e., anode), the device can remain functional. However, it is not clear from the TEM results whether the gate oxide has been ruptured at the DBIE location. For different gate oxide thicknesses and various levels of current compliance current ranges, the TEM analysis revealed three distinctive regions.³

Region I: At low compliant current and in the regime of thin gate oxides ($< 20 \text{ Å}$), SBD is rarely accompanied by DBIE.

Region II: At low or moderate compliant current and gate oxide thicker than 20 Å, SBD is always accompanied by DBIE formation.

Region III: At moderate compliance and in the regime of thin gate oxides ($< 20 \text{ Å}$), or high compliance current levels

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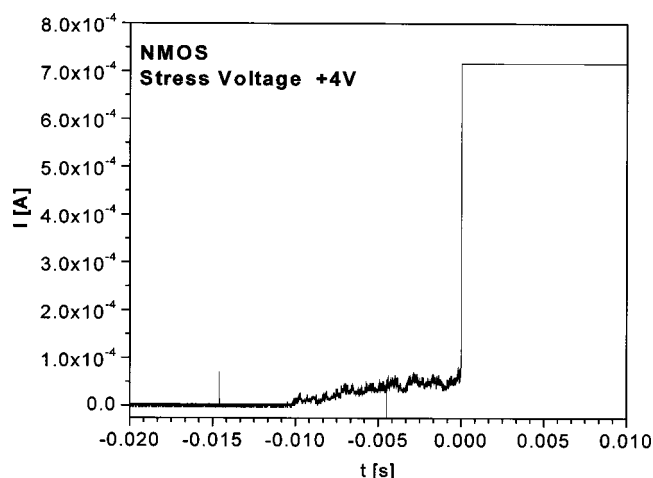


FIG. 1. Gate leakage current during a typical constant voltage stress breakdown transient of a narrow isolated transistor. The first part of the transient, typically of the order of milliseconds, is a noisy and progressive process. In the second part of the transient, the gate current jumps abruptly to very high levels in times of the order of microseconds, i.e., limited by the bandwidth of the amplifier. The gate oxide thickness is 20 Å.

regardless of gate oxide thickness, HBD is observed in most of the cases, and always accompanied by DBIE.

It is apparent from this analysis that SBD, DBIE, and HBD are at an intricate interplay during the breakdown transient. Figure 1 shows a typical BD transient of gate current, taken without applying current limit on the gate current and observed under a constant voltage stress at 4 V. The curve is the gate current during the BD transient. At least two distinctive phases can be observed from this curve. The first part of the transient, typically of the order of milliseconds, is a noisy and progressive process well in agreement with those reported recently.^{4–6} In the second part of the transient, at very high current levels, of the order of 50 μ A, the gate current jumps abruptly to very high levels in times of the order of microseconds, i.e., limited by the bandwidth of the amplifier. Figure 2 shows a typical cross-sectional TEM image of a narrow transistor after SBD. The original and post-SBD oxide/Si interfaces are delineated to illustrate the DBIE morphology. It may be noted that the poly-Si/SiO₂ interface has been pushed up by the DBIE growth on the SiO₂/Si-substrate interface.

As voltage is applied across the transistor dielectrics,

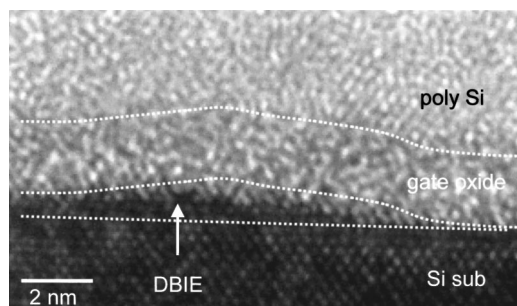


FIG. 2. DBIE nucleation bump can be seen clearly after the substrate/oxide and oxide/poly-Si interfaces delineated by the dashed line. The sample is a 20 Å gate oxide narrow *n*MOSFET stressed by CVS in inversion mode with compliance current set at 50 μ A. Electrical measurement after breakdown shows a slight degraded transistor characteristic, and the transistor is still functional, i.e., SBD.

energetic electrons “flowing” across the gate oxide will trigger several defect generation processes. The idea of damage building up in the gate dielectric to a critical level has been a key insight in formulating a predictive model of oxide breakdown. Degraeve *et al.*⁷ formulated the percolation model in which breakdown is envisioned as the formation of a connection path of defects, as a result of random defect generation throughout the insulating film. The probability of forming such a percolation path with randomly generated defects throughout the oxide bulk is computed as a function of defect density and oxide thickness. This model accounts for all the statistical features of oxide breakdown including thickness and area dependence. There is, however, not much elaboration on how the percolation path further evolves dynamically during the breakdown transient.

As a breakdown path establishes through the described percolation mechanism, a large local current density surges through the very small conductive path and generates tremendous heat around the leakage path.⁸ As most of the oxide defect generation mechanisms are thermally activated, the enhanced local thermal heating triggers almost instantaneous lateral dilation on the percolation path.

The percolation path is, thus, a dynamically evolving phenomenon during a breakdown transient. Although there is no general agreement on the nature of the defects that develop the percolation path, it has been demonstrated that the current through the path is temperature dependent,⁹ i.e., the gate leakage current increases with increasing temperature. The temperature dependence of the current in a percolation cluster is believed to be due to the scattering of electrons by a random array of scatter centers, which correspond to dangling bonds and defects present close to or within the percolation path in the SiO₂ amorphous film. At the initiation of a SBD, the initial current running through the percolation path interacts with the scattering centers and heats up the path. As the temperature rises, more current can run through the path as predicted by the percolation model.⁹ This positive feedback current–temperature loop ensures that an increasingly higher current density is pumped into the percolation path at ever increasingly higher temperature as long as the stress voltage is maintained. In the event of no other limiting factors to stop the thermal runaway, such current–temperature loop of the percolation path eventually leads to a roughly linear *I*–*V* (ohmic) characteristic, resulting in a HBD.³ This is the case as shown in Fig. 1 where *I_g* shoots up to 0.7 mA. On the other hand, it has been experimentally demonstrated that the local surge current can generate an effect similar to electrothermal migration, in which, depending on the stress polarity, Si atoms of the Si substrate or poly-Si in the vicinity of the percolation path are pushed to move in the direction of the electron flux.^{2,3} An example is shown in Fig. 2 for an *n*-type metal–oxide–semiconductor field effect transistor (*n*MOSFET) stressed in inversion mode. Eventually the dislodged Si atoms pile up at the Si/SiO₂ interface and form DBIE hillock. The DBIE formation at the Si/SiO₂ interface helps to alleviate the thermal runaway effect by consuming the system free energy and by reconfiguring the conductive path formed around the percolation path.

These two mechanisms, namely the percolation path current–temperature feedback (Fig. 1), and DBIE formation

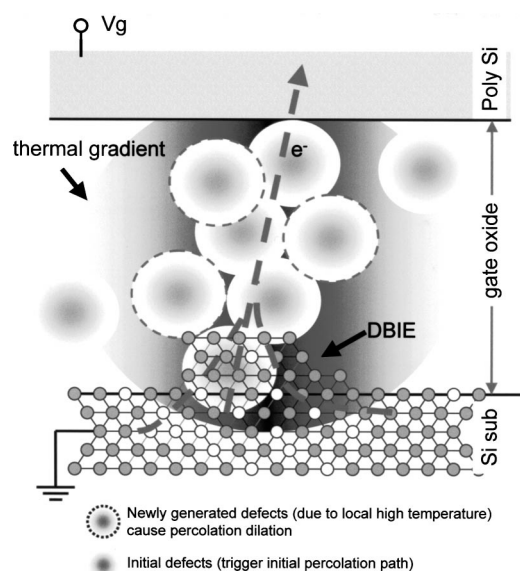


FIG. 3. During gate oxide breakdown transient, local heating increases trapped charge density near the percolation path and leads to path dilation, which allows more current to go through the percolation path. DBIE growth helps to control the thermal runaway and may prevent the SBD from evolving into HBD.

(Fig. 2), compete and complement to each other during the instance of a breakdown transient, as illustrated in Fig. 3. The following are the possibilities during a breakdown transient.

(1) In the case where there is no factor to limit the energy discharge and poor heat dissipation, the SBD current-temperature feedback first induces a DBIE. Then the occurrence of intense thermoelectronic field emission starts to kick in at a relatively high temperature of around 1500 °K. Eventually, it leads to a thermal runaway and followed by hard breakdown.^{8,10} Figure 1 shows a typical example.

(2) For very low levels of compliance current and for ultrathin gate oxides, where the stored energy and trap charge defect density are low, the current-temperature feedback is not sufficient to induce DBIE. SBD, in this case, can only sustain and maintain the percolation leakage path without inducing any physical damage to the overall transistor structure. This is the case of SBD without any DBIE.

(3) In the intermediate compliant current levels, the gate oxide breakdown event is a delicate balance between these extremes. When SBD with current-temperature feedback is sufficiently strong to trigger DBIE growth, the DBIE growth not only partially consumes the breakdown energy but also introduces local transconductance change and current fluctuation due to the generation of microstructural defects including point defects and possibly dopant redistribution. As a result, depending on the energy balance between the DBIE growth and current-temperature feedback, two potential consequences are possible.

- (a) If the current-temperature positive feedback dominates, the high current density induces a thermal runaway, leading to a HBD eventually as seen in Fig. 1; or
- (b) If the DBIE growth dominates, the current density is capped at certain level due to a negative feedback, and the system energy sustains itself without further changes, leading to a metastable equilibrium. This leads to SBD with DBIE, as shown in Fig. 2.

In summary, the percolation path has to be considered as a dynamically evolving phenomenon, in which its size and conductance are changing during the breakdown transient.^{11,12} The DBIE growth and its impact on the local channel transconductance of the transistor help to control the thermal runaway phenomenon and, in some cases, stop the SBD from evolving into a HBD. The combination of the dynamic percolation and DBIE growth is able to account for the experimental TEM observations on the gate oxide breakdown behaviors for various gate oxide thicknesses ranging from 16 to 33 Å, and for various stress polarities.

Further studies, including modeling and simulation, are needed to quantitatively understand and eventually predict this balance at various gate oxide thicknesses and stress conditions.

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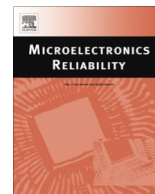
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Introductory Invited Paper

High- κ dielectric breakdown in nanoscale logic devices – Scientific insight and technology impact

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ABSTRACT

Dielectric breakdown is one of the key failure mechanisms in front-end silicon-based complementary metal oxide semiconductor (CMOS) technology. With the advent of HfO_2 -based high- κ dielectrics replacing SiO_2 and metal gate replacing polysilicon and silicides, the physics of defect generation and breakdown of the oxide has changed significantly, although the mechanisms governing operation of the transistor remain essentially the same. Given the progression towards ultra-thin dielectric films with physical thickness $\sim 1\text{--}2\text{ nm}$, the overall breakdown process has shifted from a single catastrophic hard breakdown (HBD) event to include various regimes such as soft breakdown (SBD) and progressive (post) breakdown (PBD) which in itself consists of a *digital phase* with random telegraph noise (RTN) fluctuations and stable average leakage current and an *analog phase* with gradual wear-out and lateral dilation of the percolation path resulting in a monotonic increase in leakage current. In order to better design and optimize the logic gate stack for enhancing its robustness and immunity to breakdown, it is essential to understand the driving forces and physical mechanisms behind the different phases of dielectric failure. This review is dedicated to the scientific understanding of the various regimes of breakdown in high- κ gate stacks using electrical, physical and statistical techniques along with an application of these findings to predict the impact they will have from a technology perspective.

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1. Introduction

Over the past decade, semiconductor technology has been making steady progress towards further downscaling of logic devices to keep up with the trends predicted by Moore's Law [1]. With increasing need for portable devices, higher computational complexity, increased functionality and speed (frequency), there is always a push towards making devices smaller in all aspects including length of the channel, area of the device, thickness of dielectric film, etc. As the industry headed towards the 45 nm and 32 nm technologies a few years back, the standard SiO_2/SiON dielectric posed a serious reliability bottleneck as their physical thickness was $\sim 1.2\text{ nm}$ corresponding to only 3–4 monolayers of atoms with very high intrinsic leakage current and defect density at the interfaces. This spurred the need to introduce high permittivity (high- κ) dielectrics as a replacement for SiON , so that the physical thickness of the dielectric can be increased and the equivalent oxide thickness reduced, maintaining the same coupling capacitance ($C = \epsilon_0 \cdot \kappa \cdot A/t_{\text{ox}}$) [2]. For better control of the threshold voltage (V_{th}), the gate material was also changed to metal

electrodes such as TiN and TaN with tunable work function [3,4]. Enhancement in the mobility of the channel (directly correlated to circuit speed) required a change in the substrate material with strained-Si (already in production for sub-45 nm node) [5], Ge [6] and III–V [7] materials such as InP and GaAs being considered as potential candidates for the future. In addition to the material changes, the need for further enhancement in CMOS performance has now forced designers and manufacturers to consider alterations to device architecture by replacing planar transistors with FinFETs [8] (already in production by Intel® for 22 nm node) and gate-all-around (GAA) nanowire transistors [9] that provide better channel control. With all these material and architectural evolutions taking place, the mechanism of degradation and failure of the high- κ dielectric gets increasingly complex as the breakdown phenomenon is affected by non-uniform electric field, variability at the atomic interfaces, interaction of the defect cluster and percolation path with the metal electrode and substrate as well as the role of microstructural changes in the dielectric including grain boundaries for polycrystalline films. Therefore, a detailed outlook into the different regimes of high- κ breakdown and their driving forces is essential so that the gate stack can be further optimized to reduce the defect generation probability and failure rate through changes in the process flow, material, device architecture or operation conditions or any combination of these. Fig. 1 illustrates the

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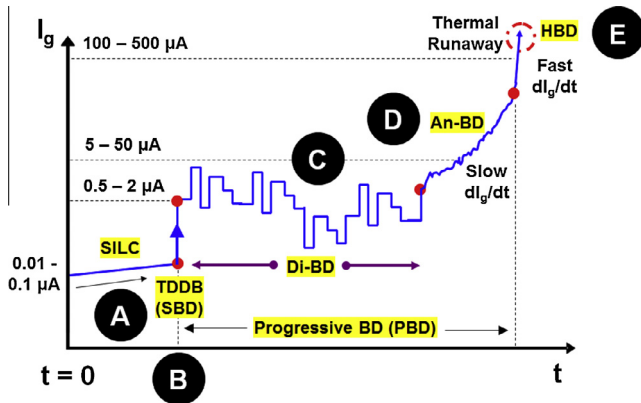


Fig. 1. General trend of gate leakage current evolution with time for a dielectric stressed at time $t = 0$. There are five regimes in the breakdown of the dielectric – (A) stress-induced leakage current (SILC), (B) time-dependent dielectric breakdown (TDDDB), (C) digital breakdown (Di-BD), (D) analog breakdown (An-BD) and (E) hard breakdown. Each of these regimes has a different kinetics of degradation. The first two regimes are device area dependent, while the rest are localized phenomena.

general trend of leakage current evolution in a gate stack starting from random defect generation prior to percolation all the way up to hard breakdown (HBD) and thermal runaway. The overall degradation of the dielectric can be categorized into five different regimes → (A) stress-induced leakage current (SILC), (B) time-dependent dielectric breakdown (TDDDB) – (this can also be referred to as soft breakdown (SBD) in ultra-thin dielectric with appropriate low compliance setting), (C) digital breakdown (Di-BD), (D) analog breakdown (An-BD) (both Di-BD and An-BD together comprise the progressive BD stage) and (E) HBD (thermal runaway destructive failure). Fig. 2 illustrates the defect density, defect spatial distribution as well as morphological changes that may be observed in these five regimes. In the next six sections, we will probe the physical phenomena underlying each of these regimes separately from an electrical, physical and/or statistical perspective and look into their dependence on the material, process, architectural and

operational aspects of the metal–oxide–semiconductor field-effect transistor (MOSFET) and also discuss the implications of these degradation mechanisms on the lifetime of the device from a technology point-of-view. Finally, this review shall conclude with a summary of the current status of know-how on high- κ breakdown and identify the various issues that still remain unaddressed and deserve further study.

2. Regime A – stress induced leakage current (SILC)

When a dielectric is stressed by voltage and/or temperature, defects are generated randomly all over the film. During this stress phase prior to any breakdown event, if the gate current, I_g (or current density) is plotted with time, we would observe a steady rise in I_g following a power law function with time (Fig. 3(a)) wherein the time exponent α can range between ~ 0.3 and 0.7 [10,11]. If the stress is interrupted at certain intervals of time and the I_g – V_g trend is measured at low voltages, the leakage curve can be seen gradually shifting upwards (Fig. 3(b)) as more and more defects are created and electrons are able to reach the gate (considering NMOS in inversion) by means of trap-assisted tunneling (TAT) [12]. This gradual and uniform degradation of the dielectric is referred to as stress-induced leakage current (SILC). The shift in I_g – V_g trend can be much larger closer to the instant of breakdown during SILC because the probability of some defects overlapping with each other and forming defect clusters increases thereby resulting in quantum jump of conductivity (higher tunnel probability) (Fig. 3(b and c)). As more defects nucleate and some are closer to each other, the effective overall tunneling distance reduces significantly, thereby enhancing the tunnel current significantly. Note that the tunneling probability is exponentially dependent on the barrier thickness, as governed by the Wentzel–Kramers–Brillouin (WKB) approximation [13]. There are different schools of thought on the SILC model for fitting the I_g – V_g data of any given device. The popular models include elastic TAT [14], inelastic TAT with the carriers believed to lose energy in the defect site in the form of phonons [15] and Poole–Frenkel emission [16] where carriers

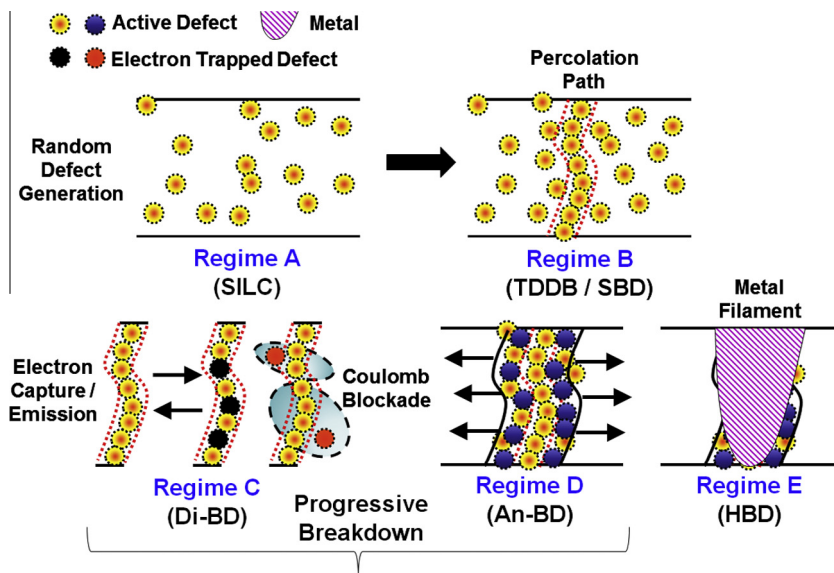


Fig. 2. Illustration of the defect density and spatial distribution in the dielectric corresponding to the five regimes of breakdown shown in Fig. 1. The yellow and blue dots represent active defects in the pre-TDDDB and post-TDDDB stage. The black and red dots correspond to the state of trapped charge. It is believed that defects that are trapped with electrons cannot assist in current transport as effectively compared to empty defects (unfilled energy states). All the regimes except the last one (HBD) involve the role of oxygen vacancy defects. The HBD stage is characterized by migration of metal from the gate electrode and/or silicon from the substrate. The digital BD regime involves random telegraph noise signals due to stochastic electron/capture emission events at the defect sites and/or the Coulomb Blockade effect due to traps in the vicinity of the percolation path that alter its conductance. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

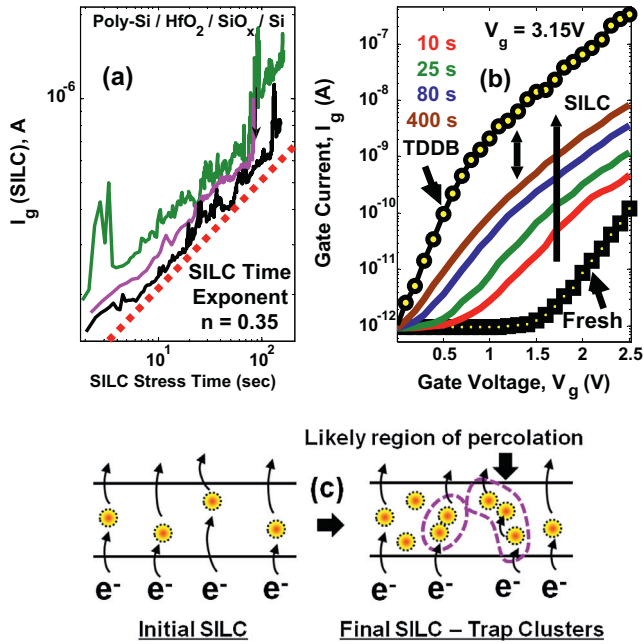


Fig. 3. (a) Time-dependent power law trend of gate current evolution, prior to percolation breakdown, plotted on a logarithmic scale. (b) Trend of sensed I_g - V_g data for a device with the stressing at $V_g = 3.15$ V halted at different time instants of 10 s, 25 s, 80 s and 400 s. (c) Illustration showing the clustering of defects (shown by dashed contours) during the final stages of SILC prior to breakdown that reduce the local barrier thickness significantly causing the leakage to jump by more than an order of magnitude as the test data in (b) shows.

hop (not tunnel) across the potential barriers surrounding the trap by thermal energy. Although there is no consensus on which of these models is the best one, the inelastic TAT approach is the most generic framework [15,17,18] that works well for I_g - V_g fitting of transistors across various CMOS technology nodes (including Flash memory [15] and resistive switching memory [19] as well). Apart from the conduction models, there are different mechanisms proposed for defect creation in the dielectric including anode hole injection (AHI) (charge-fluence based) [20], thermochemical model (electric field based) [21] and hydrogen release model (interface phenomenon) [22]. Each of these models have their own pros and cons and the most suitable model has to be chosen depending on the gate stack (materials), majority carriers (electrons/holes) and stress conditions. In most cases, no single model describes all the observed degradation phenomenon and therefore, a convolution of these models may provide a better insight to the actual physics of defect generation in high- κ dielectric films.

The kinetics of defect generation during SILC very much depend on the microstructure of the dielectric. While SiO_2/SiON always remain amorphous, hafnium-based high- κ stacks have a moderate crystallization temperature (~ 400 – 500 °C for HfO_2 [23]) that can cause the dielectric to become polycrystalline in nature with grain boundary fault lines (dislocations) either during the deposition or the annealing process. Since grain boundaries already contain pre-existing defects (process-induced), they serve as weakest link paths and defect generation is more localized along these GB regions [24–26]. As a result, SILC is no longer random, but a localized phenomenon around the GB. Moreover, atomistic studies also reveal that oxygen vacancies (defects) have a thermodynamic drive towards segregating around the GB region (Fig. 4) [27]. The typical size of the grain can be ~ 20 – 25 nm [28], while the diameter of the GB is as low as 4–6 Å [27]. For a given transistor with dimensions say 70 nm $\times$ 100 nm, we can expect to see many GB paths where SILC is more predominant. Analysis of polycrystalline dielectric thin films using conductive atomic force microscopy (CAFM)

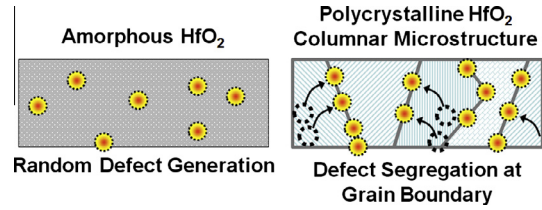


Fig. 4. Defect generation kinetics in amorphous and polycrystalline dielectric films with spatially random and localized trends respectively. Even defects generated in the grain (bulk) of the polycrystalline films tend to migrate (as indicated by the arrows) and segregate at the grain boundary so as to minimize the overall system free energy.

provides clear evidence that the initial leakage is higher at GB locations and subsequent defect generation also localized around these regions, as shown by one such CAFM scan micrograph in Fig. 5 after repeated stress scans [29].

Recent reports by Cartier and Kerber [30] suggest that SILC is a recoverable phenomenon in high- κ gate stacks when a negative polarity voltage is applied (after positive bias stressing, i.e. a bipolar voltage scheme). With electrical studies showing these trends, the physical mechanism of SILC recovery was initially attributed to charge trapping at the defect sites [30]. However, since trapping and detrapping are stochastic phenomenon and cannot cause a ‘permanent’ recovery, we hypothesized that the recovery is possibly due to the recombination (passivation) of the oxygen vacancy defects with the oxygen ions that were initially liberated from the dielectric during the stress phase and stored in the metal gate (MG) which serves as a good oxygen reservoir [31].

The degradation in V_{th} and saturation drain current (I_{D-SAT}) during SILC and the rejuvenation of these parameters during the recovery stage (by applying the opposite polarity voltage) is shown in Fig. 6(a and b) for a single cycle [31]. As shown by Fig. 6(c and d), with higher pre-breakdown compliance setting (from 100 nA to 1 μ A), the degradation in I_{D-SAT} is higher while the extent of recovery is accordingly lower [31]. In other words, it becomes increasingly difficult to repair the defects as the dielectric continues to degrade approaching the percolation point. Nevertheless, the stable recovery of SILC leakage can be a practically very useful concept because a simple reverse polarity voltage sweep across a circuit (‘reflash’) can in fact help to repair some of the process or stress induced defects and enable lower power dissipation. This, in layman terms, could mean that the electronic gadget can be prevented from getting too hot in the first few years of its operation.

3. Regime B – time dependent dielectric breakdown (TDDB)

As the SILC stage continues with more and more defect clusters being formed, we reach a stage when one particular combination of defects form a chain linking the gate to the substrate. This phenomenon is called “percolation” [32] and the mechanism is referred to as *time-dependent dielectric breakdown* (TDDB). The current jump at this instant is catastrophic and the extent of damage to the dielectric is only controlled by the external compliance setting (I_{comp}) using the parameter analyzer. For thick dielectrics with $t_{ox} > 4$ – 5 nm in the past, the TDDB was the endpoint for oxide failure as the percolation event itself induced extensive damage that caused transistor malfunction and rendered the circuit unfit for further operation. However carefully the I_{comp} value was chosen, the compliance overshoot effect (which may be in the nanoseconds range and faster than the maximum operational frequency of the parameter analyzer) could not be overcome. On the contrary, when thin dielectric films of $t_{ox} < 2$ – 3 nm are considered, the TDDB event can be very well controlled and the breakdown is essentially “soft” in nature. The reason behind this observation is illustrated by

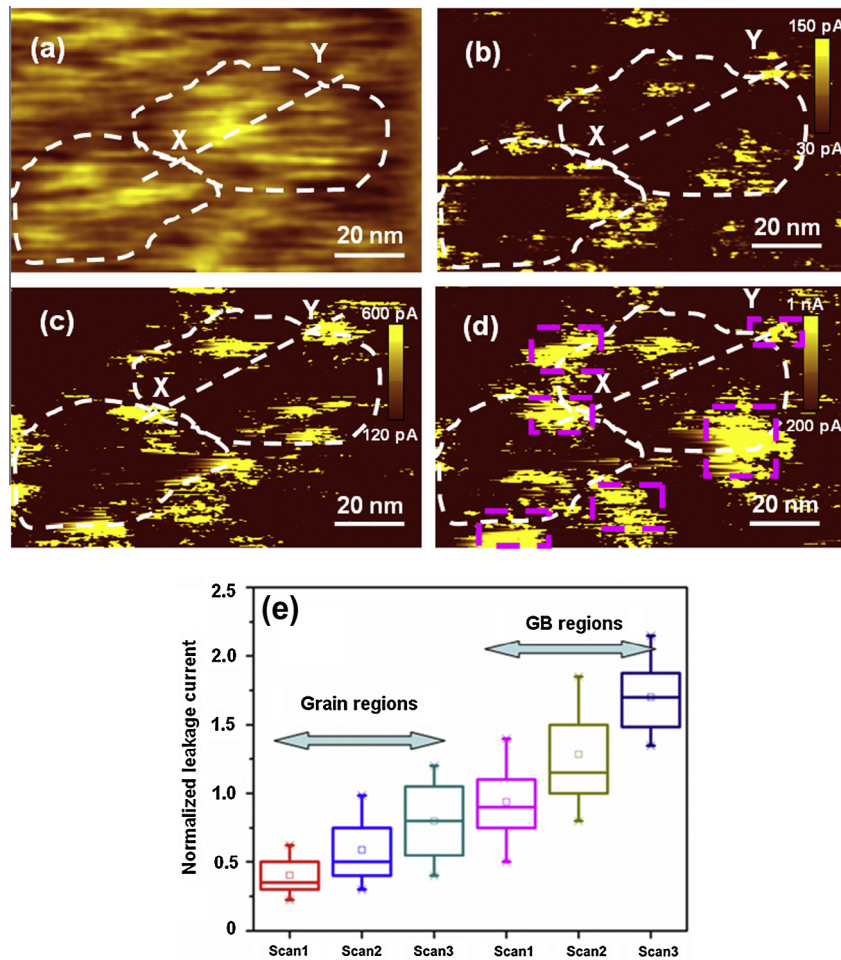


Fig. 5. (a) CAFM topography, leakage current profile after the (b) 1st scan, (c) 2nd scan and (d) 3rd scan [29]. (e) Boxplot of the normalized leakage current distribution at the grain and GB regions after each successive scan. It can be observed that defect generation is clearly localized around the GB contours (indicated by the white dotted lines).

Fig. 7, wherein for a thick dielectric, a lot of defects are created all over the film (oxide becomes highly defective) before the percolation probability is high enough for TDDB to occur. Since there are many defects surrounding the percolation path, the wear-out process is significant and harsh with a positive feedback of the localized temperature in the percolation path aggravating further defect generation which in turn increases the percolation path current density and so on. For thin dielectrics though, very few defects would have nucleated at the vicinity of the percolation path thereby curbing the wear-out kinetics sufficiently.

The shift in the I_g - V_g trend after TDDB can be seen in Fig. 3(b), where the leakage current is now around three orders of magnitude higher than in the fresh device. As compared to the SiO_2/SiON based technology node, the TDDB trends look very different in the high- κ stack because there is still a thin native SiO_x ($x < 2$) interfacial layer (IL) in between the high- κ and silicon substrate, which makes it a bi-layer film, each with different dielectric strength. A carefully executed TDDB test with appropriate choice of voltage stress and I_{comp} enables us to visualize this two-stage TDDB trend [33] in the HK-IL stack, as illustrated in Fig. 8(a). There is at least an order or two difference in the magnitude of the leakage current depending on whether both the dielectric layers broke down or just one of them (Fig. 8(b)). Although there still exists a lot of controversy as to which layer fails first, with some groups suggesting HK to be the first to fail [34–36] and others suggesting IL to be the one [37–42], our analysis considering the critical field strength of SiO_x and HfO_2 and the defect generation rate in both layers, points to IL being the weakest link [43,44].

Lifetime data for TDDB failure is generally described by the Weibull distribution [45], which has the weakest-link property. However, most TDDB test data on HK stacks showed significant deviations from the Weibull trend especially at low percentiles. Our statistical cumulative damage model [33] which accounted for the time varying step-stress profile across the second dielectric, after the first one percolates was able to successfully explain and model the non-Weibullian behavior. It becomes critical therefore to use the sophisticated model that best represents the kinetics of the breakdown phenomenon, when we want to assess the lifetime distribution and extrapolate the field use lifetime of HK-based stacks.

The vulnerability of the GB region to undergo faster TDDB is evident in Fig. 9 using the scanning tunneling microscopy (STM) setup with stressing carried out on blanket CeO_2 -based films [46]. It is detrimental to have GB in the processed stack, and the crystallization process in turn depends on the dielectric thickness and annealing conditions. The non-uniform local defect generation in and around GB was also modeled using percolation theory and we proved that the presence of GB can also result in convexial TDDB distribution trends [47]. Note that the Weibull Slope is an important parameter for dielectric breakdown analysis as it is directly proportional to the number of defects needed to form the percolation path and it determines the variability of the TDDB distribution. Thinner dielectrics will have lower β (wider spread in time to failure) and vice versa.

When the MG-HK transistor after TDDB using low compliance capping (corresponding to soft breakdown) is analysed using high

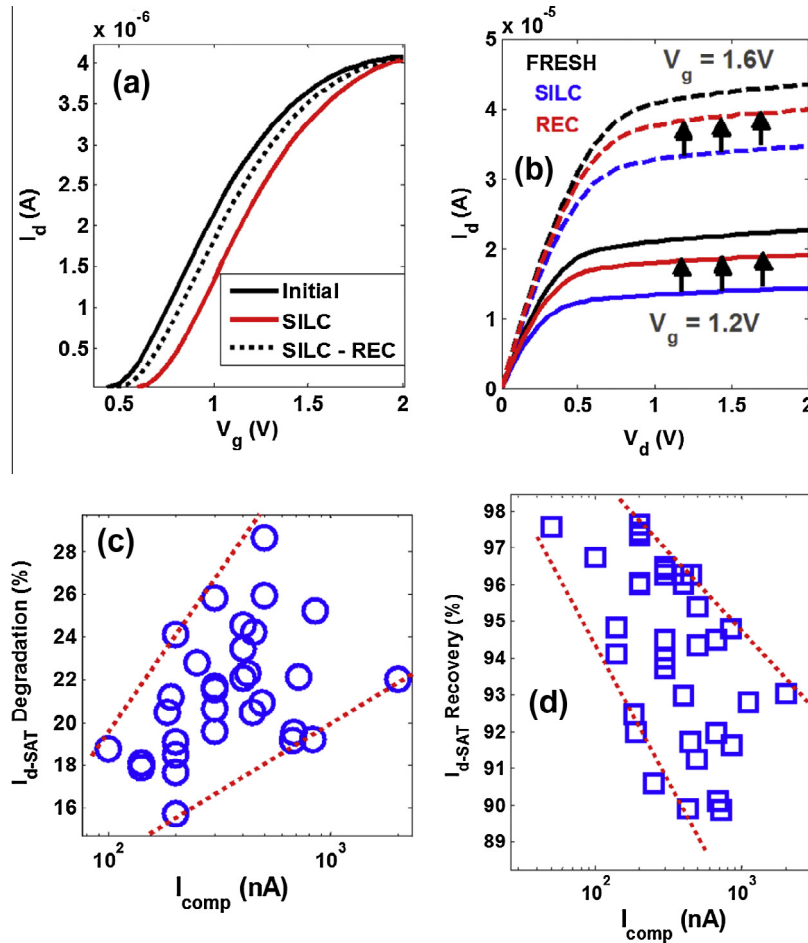


Fig. 6. Recovery in (a) V_{th} and (b) I_{D-SAT} for a single cycle of SILC with positive voltage stress followed by reverse bias I - V sweep. The dependence of the degradation and recovery in I_{D-SAT} as a function of the current compliance level is plotted in (c) and (d) for many devices and many stress-recovery cycles [31].

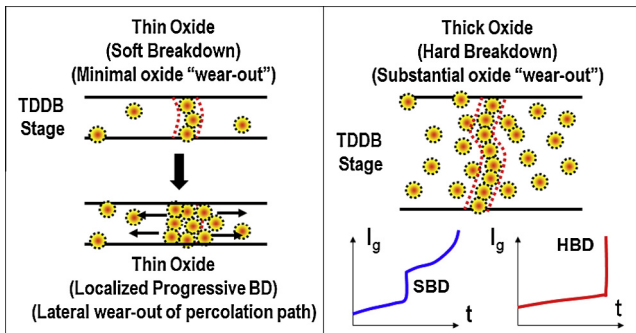


Fig. 7. Illustrating the reasoning behind the high likelihood of HBD in thick dielectrics due to the large defect density surrounding the region where percolation is about to take place. The transient of the percolation event is difficult to control for thick films. On the contrary, the ultra-thin films show a very soft BD event, followed by a gradual wear-out process to HBD conditions. The corresponding I_g - t curves for SBD and HBD are also shown.

resolution transmission electron microscopy (HRTEM), there is no clear physical or morphological change observed at the failure site (Fig. 10(a and b)) [48]. Note that the failure site along the length of the channel is estimated by calculating the weighted ratio of the source and drain currents, as suggested by Degraeve et al. [49]. When the failure site is examined by electron energy loss spectroscopy (EELS) (Fig. 10(c)) [50] for oxygen content, a lower oxygen count at the percolation path is confirmed, when compared with

a reference region of the dielectric which did not degrade much during the stress. This suggests that the primary defect or trap is an “oxygen vacancy” and it is generated due to the field induced thermochemical bond breakage of the Hf–O or Si–O bond.

The performance of the transistor in terms of its transfer curve (I_d - V_g) and I_d - V_d characteristic are degraded after SBD, but the functional form still remains intact. This implies that the device and circuit can still function normally in the event of a SBD (Fig. 11). As was the case for SILC, it is also possible to achieve partial recovery of the damage caused by the SBD phenomenon in metal gate – high- κ stacks, when a moderate reverse bias I - V sweep is applied (Fig. 12) [51]. The extent of recovery depends very much on the oxygen solubility of the gate electrode. Investigation of different electrode materials (Ni, Ta, Ti and Si) reveal that recovery is best seen in Ti with 33 at.% and lowest in Si with 0.004 at.% oxygen solubility [51]. The term ‘oxygen solubility’ here refers to the ability of the metal electrode to store O^{2-} ions (which are generated when the Hf–O/Si–O bond is broken) in mobile form, without getting oxidized.

4. Regime C – digital breakdown (Di-BD)

If the dielectric has suffered a SBD event and we continue to stress the dielectric with a certain gate voltage, we observe discrete and random jumps in the current levels corresponding to random telegraph noise (RTN). These RTN signals arise from the stochastic capture and emission of electrons into and out of the vacancy defect sites (traps) constituting the percolation path. A trap filled

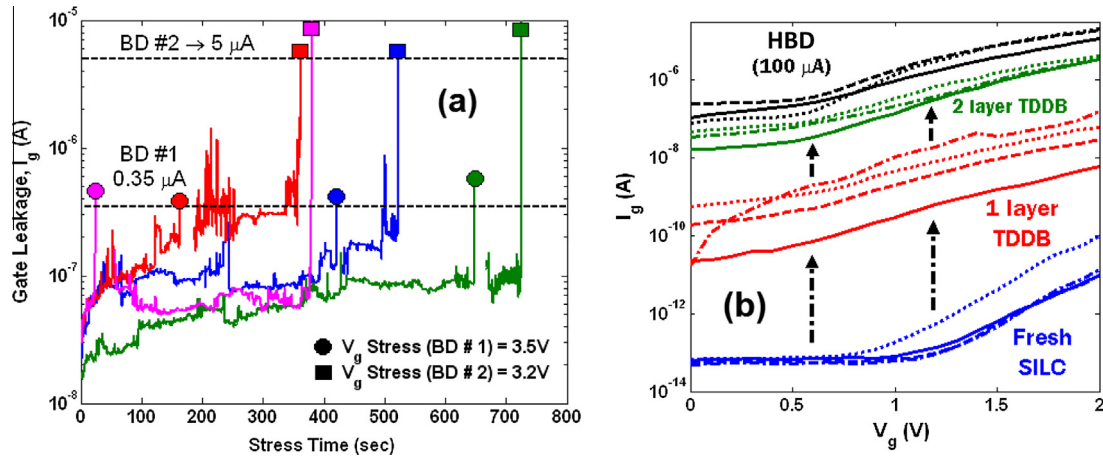


Fig. 8. (a) Detection of a two-stage TDDB breakdown trend in the dual layer HK–IL gate stack for four similar devices. The end of the first and second stages of TDDB are denoted by the circle and square symbols respectively. Observation of such stage-wise breakdown requires very precise control over the compliance and stress condition. Here, we chose a very low compliance of 0.35 μA to quickly arrest the breakdown of the first layer, following which a much lower stress voltage is used to percolate the second layer in finite time [33]. (b) Trend of I_g – V_g for fresh device, 1-layer BD, 2-layer BD as well as progressive BD with significant wear-out of percolation path. There are clear jumps in the I – V trend depending on the number of layers that percolated.

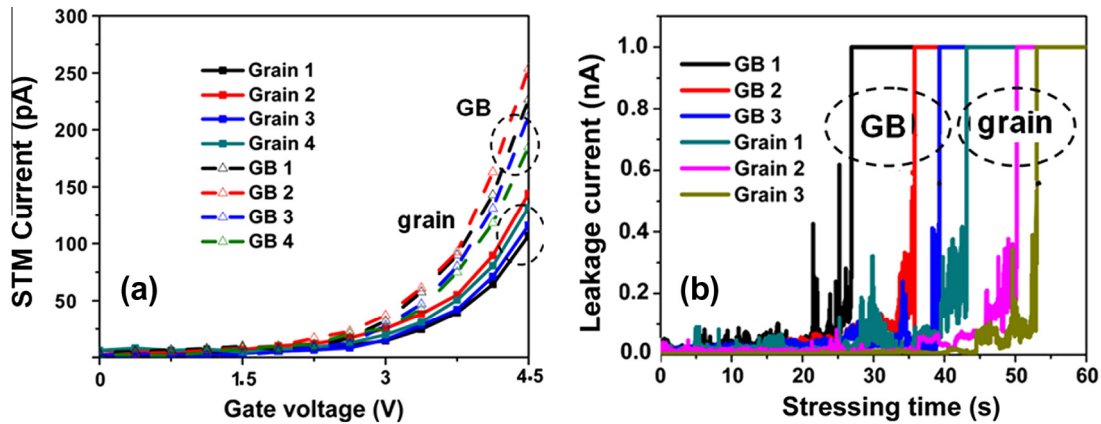


Fig. 9. (a) STM I – V scan of four grain and GB locations in the polycrystalline dielectric (the material is CeO_2 in this analysis). The GB regions have a much higher conductance. (b) Localized TDDB stress at the grain and GB sites reveals that GB tends to percolate much earlier due to the higher density of pre-existing defects [46].

with electron has the lowest ability to allow other electrons to tunnel through and vice versa. While the current fluctuates at 2^N levels if we have N traps [52], the mean value of the leakage current does not change. Therefore, the performance of the transistor in the Di-BD stage can range from being close to a fresh device if all the traps in the percolation path are occupied with electrons or being close to the TDDB stage when all the traps are active and empty. Di-BD is a safe region to continue operating the transistor in; however, note that the Di-BD phase does not stay on forever, but gives way gradually to the wear-out process, which will be detailed in the next section. Fig. 13 shows the RTN trends observed after SBD in SiON and high- κ gate stacks. The RTN pattern for the HK stack is relatively noisy due to the additional contribution of traps at other non-percolated locations (GB or interface traps). For multi-level RTN, Fig. 14 is a very good example which clearly shows the 16 current levels corresponding to $\log_2(16) = 4$ active defects in the percolation path [52].

5. REGIME D – analog breakdown (An-BD)

In the digital regime, when the current exhibits an RTN pattern with current ranging between 10^{-9} and 10^{-7} A flowing through a small percolation path ~ 2 – 5 nm in size, the current density (j_{perc})

can be moderately high. If one additional defect is created close to the percolation path, the value of j_{perc} marginally goes up, leading to enhanced Joule Heating and localized rise in the temperature of the percolation path (T_{perc}). The rise in T_{perc} enhances the probability of subsequent vacancy generation which again causes j_{perc} and T_{perc} to rise and so on. This positive feedback mechanism eventually causes the percolation path to dilate laterally or the oxide to thin down vertically [53]. In other words, the percolation path enters the ‘wear-out’ phase and the current increases monotonically (with RTN effects appearing subdued) with time, as shown in Fig. 15. The evidence for lateral dilation of the percolation path comes from the EELS mapping of the breakdown spot for a wide range of I_{comp} from as low as 2 μA to as high as 40 μA [54] (Fig. 16). If we visualize the percolation path to be cylindrical in shape, we can estimate that the oxygen deficiency distribution broadens out from a radius of 2–5 nm all the way up to 30–40 nm as I_{comp} is increased and the central core becomes increasingly metal/semiconductor-like due to almost complete depletion of oxygen (Fig. 17(a)). The other possibility of oxide thinning was also observed using HRTEM micrograph [53] (Fig. 17(b)) where the crystallographic Si from the substrate was observed to gradually protrude into the dielectric (maintaining its crystal orientation), causing local effective oxide thinning (forming a silicon

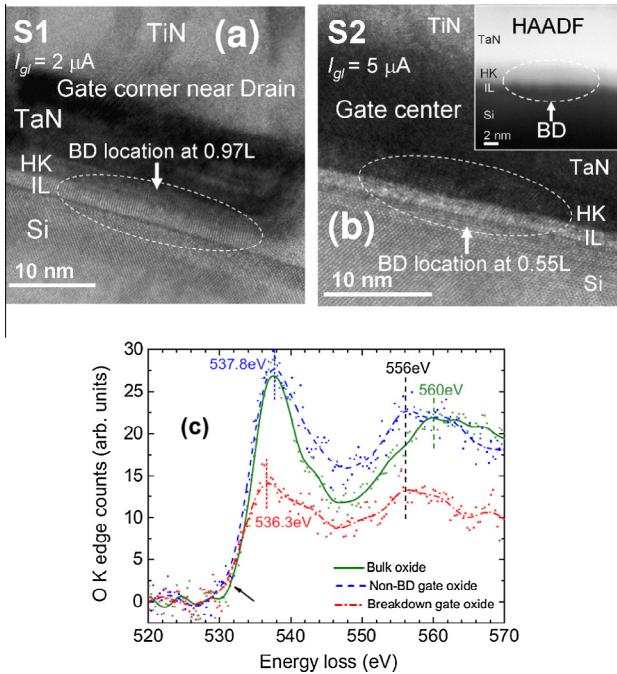


Fig. 10. (a and b) HRTEM micrograph of two devices subjected to TDDDB with low compliance of 2–5 μA corresponding to SBD. The location of the BD plot is also labeled in the figure with OL and IL corresponding to source and drain corners respectively. (c) EELS analysis of O K-edge count in one such device shows a consistent drop in the oxygen signal (red line) when compared to a reference region where not many defects were nucleated (blue line) [50]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

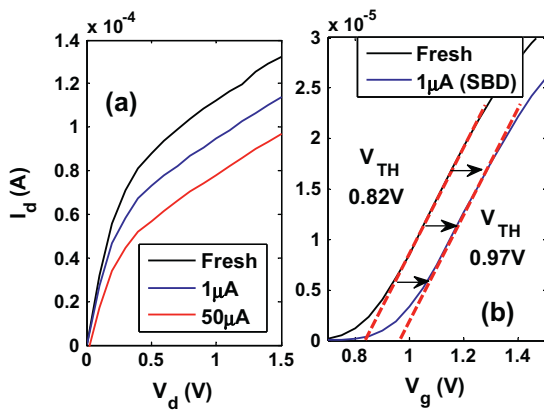


Fig. 11. Plot of the (a) I_d – V_d and (b) I_d – V_g characteristic of a MG–HK device comparing the virgin (fresh) trend with that after soft breakdown (1 μA). Clearly, while the performance has degraded (lower saturation drain current and higher V_{TH}), the transistor functionality still remains intact. As an illustration, further degradation in drain current with wear-out of the percolation path is shown in (a) for $I_{\text{comp}} \sim 50 \mu\text{A}$.

hillock), possibly due to thermo-migration and high current density induced electron wind force [53]. This defect is known as ‘dielectric breakdown-induced epitaxy (DBIE)’. The size of the DBIE protrusion increases with higher I_{comp} , eventually shorting the gate to the substrate.

If the transistor performance in the Di-BD and An-BD phases is compared (Fig. 18), the transfer and drive current characteristics for An-BD are significantly degraded with increasing wear-out and are deemed unsuitable for a reliable functional operation of the circuit. Therefore, the end of useful life of the device should

be defined as the juncture when the device transits from the Di-BD to An-BD phase.

6. Regime E – hard breakdown (thermal runaway)

The most destructive phase of breakdown is HBD (also referred to as thermal runaway) and typically occurs when the TDDDB ultra-fast transient is not well controlled or a low-enough compliance cap is not chosen when stressing the metal gate stack. As shown by the HRTEM micrographs for Ta and Ni gated transistors in Fig. 19(a and b) [55,56] and Fig. 20(a) [57], the HBD stage involves migration of the metal atoms/ions from the gate into the dielectric and protruding all the way to the substrate. We call this as ‘punch-through’ or ‘filamentation’. The pattern of filamentation in Ni and Ta stacks is drastically different. While Ni forms spikes with diameter as low as 2 nm, Ta forms bowl-shaped filament with size ~ 15 –20 nm. It appears that Ni diffuses more directionally along [111], while Ta shows more isotropic transport. The tendency of Ni to spike along the [111] direction has been well reported in process reliability studies for fully-silicided (FUSI) high- κ gate stacks [58,59]. We confirm that the morphological patterns observed indeed correspond to Ta and Ni atoms using energy dispersive X-ray (EDX) and EELS analysis in Fig. 19(c) and Fig. 20(b) respectively [55,57]. If the I – V curves in the HBD phase were measured, we would observe complete loss of transistive action, as illustrated by Fig. 21.

If the polysilicon gate based transistor is intentionally driven to the HBD regime, we observe a severe DBIE effect that shorts the gate and substrate completely or a fully O-depleted silicon or hafnium nanowire region, resulting in resistive (Ohmic) conduction [53]. Both these defect signatures are shown in Fig. 22(a and b). Also, it is possible that the silicide contacts to the source and drain may migrate through the defective interfaces and enter the channel region, leading to a ‘channel short’ in the extreme case. This is referred to as dielectric breakdown induced metal migration (DBIM) [57] and Fig. 22(c) illustrates this case. Table 1 summarizes the different defect signatures (and their driving forces) observed for different CMOS technology generations (polysilicon, NiSi-based and TaN/TiN-based) in the HBD mode. The analysis so far has looked at breakdown from a scientific viewpoint trying to understand the intricate details of the various failure mechanisms, their root cause and material dependencies. In the next section, we will evaluate the technology implications of our understanding on high- κ dielectric breakdown and attempt to propose design for reliability (DFR) initiatives to help enhance the front-end robustness of future CMOS gate stacks.

7. Technology implications of high- κ breakdown study

- To investigate the behavior of the transistor in the five different regimes so far, we intentionally stressed the device severely to reach the desired state and extent of damage. However, for a real integrated circuit, operating at $V_{\text{op}} = 1 \text{ V}$, is it feasible to observe all these five regimes? It turns out that first of all, the high- κ dielectric is so robust (high bond activation energy for $\text{HfO}_2 \sim 4.4 \text{ eV}$ [60]) that even after the IL layer breaks down, the trap generation rate in the HK layer (even with increased local stress) is still so low that the HK seldom breaks down [43]. This means that in a realistic time span of 10 years, we may not observe a full percolation path connecting the gate and substrate for $V_{\text{op}} = 1 \text{ V}$ in a dual layer gate stack. In the absence of a full percolation, the current density is too low to cause any severe wear-out effects within the IL layer. This implies that the only way the circuit could fail (standard circuit failure criterion in the industry $\rightarrow 10 \mu\text{A}$ at $V_{\text{op}} = 1 \text{ V}$) is due to

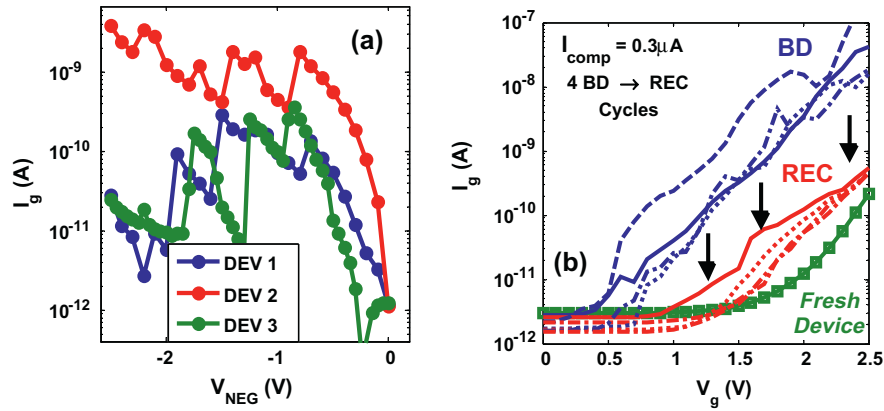


Fig. 12. Recovery trend of gate leakage after SBD event with $I_{comp} \sim 0.3\text{--}1\text{ }\mu\text{A}$ [51]. The plot in (a) shows the actual recovery trend during the negative $I\text{--}V$ sweep for three different devices with discrete jumps representing the passivation of oxygen vacancies by oxygen ions that drift from the gate electrode. In (b), the $I_g\text{--}V_g$ after SBD and after recovery are plotted for many cycles in the same device. The recovery can be performed many times after SBD, but it is seldom possible to return back to the virgin $I\text{--}V$ data as the probability of all the traps being passivated is very low.

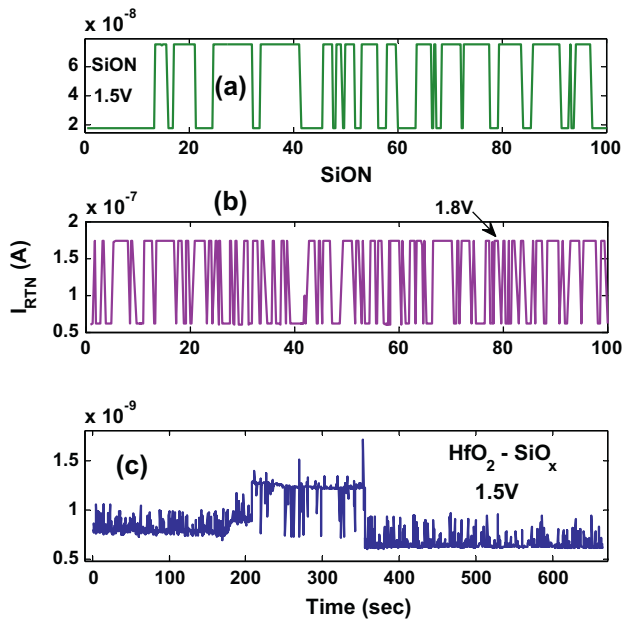


Fig. 13. RTN trends measured on (a and b) SiON at $V_g = 1.5\text{ V}$, 1.8 V and (c) HK–IL stack at $V_g = 1.5\text{ V}$ after soft breakdown at $I_{comp} = 1\text{ }\mu\text{A}$. In this Di-BD regime, the current keeps fluctuating about a mean value. Transistor performance in this regime can still be quite good, though variability in performance characteristics may be large.

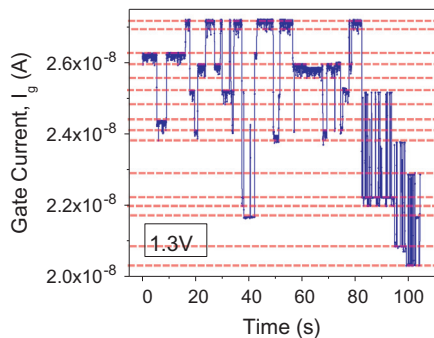


Fig. 14. Multi-level RTN signal observed after SBD in a TiN/HfLaO ($18\text{ }\text{\AA}$) – p-Si device with area of $500 \times 70\text{ nm}^2$ and $I_{comp} = 2\text{ }\mu\text{A}$ [52]. We can clearly identify 16 levels of current in the signal which corresponds to 4 active defects comprising the percolation path.

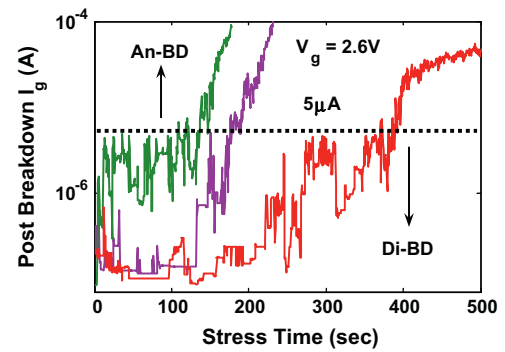


Fig. 15. Gradual evolution of the gate leakage current from the Di-BD regime to An-BD regime with the fluctuations becoming less visible as the wear-out kinetics of percolation path start to dominate the leakage and power dissipation. We show three devices consistently showing these transition trends at $V_g = 2.6\text{ V}$.

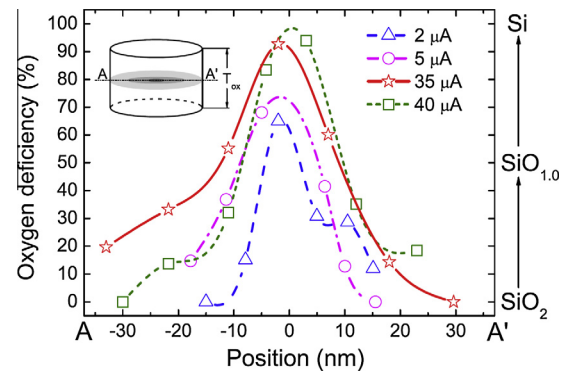


Fig. 16. EELS spectroscopy of relative oxygen deficiency in the percolation path and the region surrounding it as a function of different I_{comp} values corresponding to varying degree of wear-out and breakdown hardness in a SiON MOS device [54]. The roughly Gaussian distribution trend gets wider with higher core peak as I_{comp} is raised. This implies that the percolation path laterally dilates and the central core of the percolation becomes more silicon-like with harder breakdown and severe wear-out.

multiple correlated/uncorrelated SBD events within the IL layer [61]. In short, the SILC and TDDDB within the IL layer are the only two regimes that govern the circuit reliability of current CMOS technology. We have verified our hypothesis here using compact modeling simulations (making use of the phonon trap-assisted tunneling (PTAT) model [62] and thermochemical

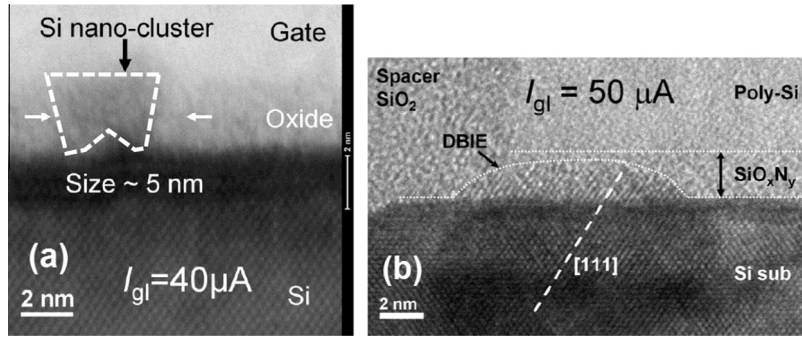


Fig. 17. (a) Observation of a silicon nanowire (nanocluster, marked by white dotted lines) with darker contrast in the core of the percolation path due to almost complete depletion of oxygen and (b) protrusion of silicon epitaxy (DBIE) into the oxide region causing effective oxide thinning. Both these micrographs are taken from the analog breakdown regime ($I_{comp} \sim 40\text{--}50 \mu\text{A}$) of SiON device technology [53]. Similar results are expected for moderate compliance levels, if we were stressing the HK technology as well.

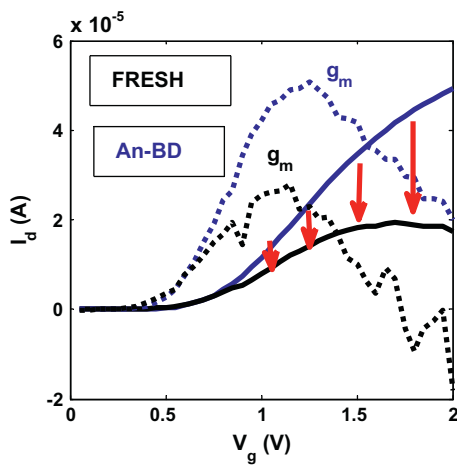


Fig. 18. Plot of the I_d – V_g curve for fresh device and after An-BD ($I_{comp} \sim 30 \mu\text{A}$). There is a significant drop in the current and the shape of the curve has also changed. The dotted lines represent the transconductance plot.

defect generation model [63] of defect generation in dual layer stacks right from SILC all the way up to An-BD. Refer to Fig. 23 for the results of the simulation [61]. Given the four possible scenarios of post-breakdown in Fig. 24 [61], Scenario C (multiple IL SBD events) is the found to be the most realistic mechanism that can explain circuit degradation under operating conditions. Scenario A (wear-out of IL percolation path) plays a secondary role, while Scenario B (full bi-layer percolation) and D (metal filamentation) are completely ruled out. Note that while the GB in the high- κ serves to increase the field locally in the IL layer beneath it, the GB itself seldom breaks down in the dual layer stack. The sub-stoichiometry of the IL layer (SiO_x , $x < 2$) determines its permittivity ($\kappa \sim 6\text{--}9$ for SiO_x , $\kappa \sim 3.9$ for SiO_2) [64] and this plays a critical role in the field distribution across the HK–IL (using Gauss Law [65]) which in turn determines the circuit level reliability at $V_{op} = 1 \text{ V}$.

- The statistical estimation of field use lifetime of a transistor comes with large uncertainty due to three main reasons. One is the highly accelerated stress conditions used for short duration reliability tests where it is uncertain whether the failure mechanism being accelerated is the same as that we would have observed at $V_{op} = 1 \text{ V}$ [66]. The second is the insufficient knowledge on the physical mechanism of defect generation and breakdown as well as the questionable validity of the standard Weibull distribution [67] (this is more so for new material stacks) and the last one is the empirical extrapolation model, which can yield lifetime estimates at V_{op} that vary by more than

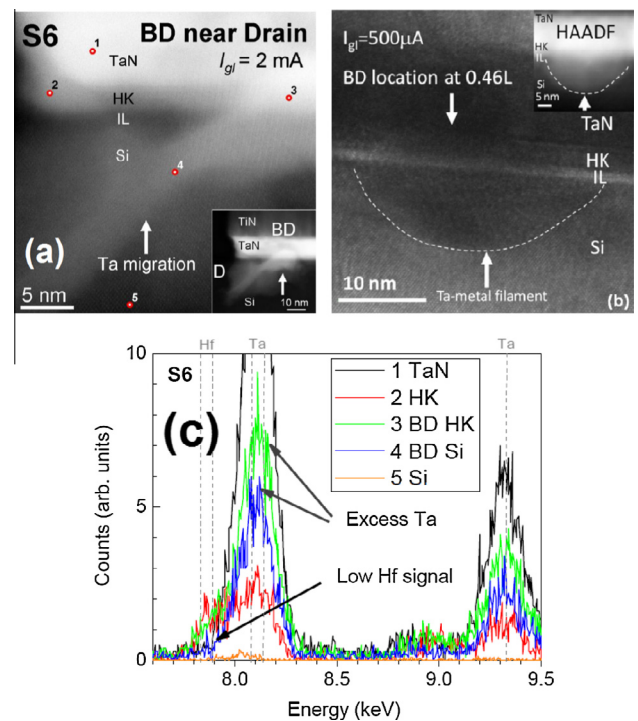


Fig. 19. Observation of filamentation in the TaN gate stack when HBD is triggered using a high (a) $I_{comp} \sim 2 \text{ mA}$ [55] and (b) $I_{comp} \sim 500 \mu\text{A}$ [56]. The bright contrast protrusion in (a) is confirmed to be Ta by using the EDX analysis as shown in (c), where EDX scan is performed at five different locations (labeled in (a)) [55]. Location 3 in the dielectric shows a high Ta signal, which is a clear indicator of metal filament formation.

5–10 orders of magnitude depending on the power law or exponential expressions that are considered [68]. For quantitative reliability assessment of a new technology generation, it is necessary to model the failure probability starting from the basics rather than crudely assuming the Weibull model to hold true all the time. As already discussed, the TDDDB phenomenon in polycrystalline high- κ and dual layer HK–IL stacks has been proven to be non-Weibullian in nature [47,33]. If we begin the reliability assessment exercise with a wrong distribution or model or inappropriate test conditions, then the effort and resources spent get wasted.

- Most of the standard TDDDB tests in the industry aim to have a large sample set of data within a short span of time and this is always executed in a standard fashion keeping the I_{comp} high at $\sim 100 \mu\text{A}$. It is actually erroneous to follow these standard test

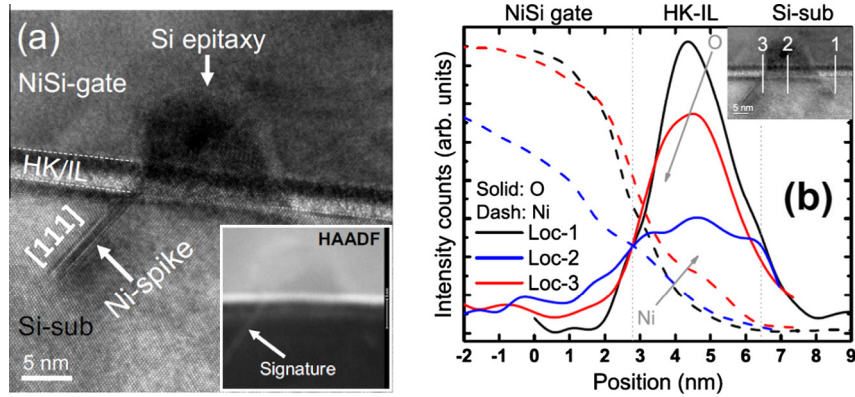


Fig. 20. (a) Observation of Ni spiking into the dielectric and the silicon in [111] direction during HBD at $I_{comp} \sim 500 \mu A$. The inset shows the high angle annular dark field (HAADF) image of the same area. (b) Using EELS analysis at three different locations (labelled in inset), we confirm the presence of Ni inside the dielectric at the location where the spike is seen in (a) [57].

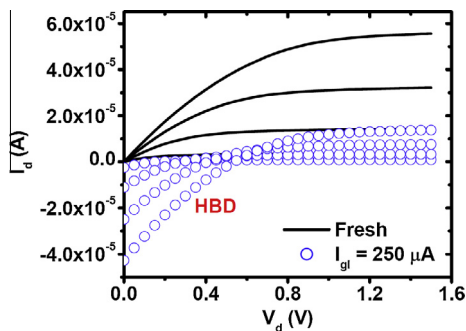


Fig. 21. Trend of I_d – V_d after HBD of the dielectric at $I_{comp} \sim 250 \mu A$. The transistor loses its functionality completely.

patterns all the time, more so, in the case of HK–IL dual layer dielectric films because the stress levels being used are so high that once the first dielectric breaks down (say IL), the second one would catastrophically break down immediately as it experiences a sudden surge in the local ξ -field, that is much beyond its critical field strength. As a result, the second dielectric does not undergo a proper reliability test as the intrinsic degradation kinetics was never studied (rather we overstressed the sample). Therefore, design of electrical test algorithms for intrinsic TDDB lifetime analysis has to be carefully done. Most industrial research groups have to reconsider their standard reliability test routines and modify them to measure and analyze the two-stage finite time breakdown phenomenon. We have recently proposed an algorithm in Ref. [33] to successfully decode the

two stages of TDDB by carefully controlling the stress voltage and the compliance needed to arrest the percolation transient effectively.

- To continue aggressive downscaling of device dimensions, the proposal of a zero interfacial layer (ZIL) stack is being considered for future sub-16 nm nodes [69]. In the absence of the IL, the HK sits directly on the silicon and the GB links (kind of ‘short circuit’) the gate and substrate leading to high leakage currents and poor TDDB robustness. In such cases, a two-stage HK film deposition may be useful [70] so that the GB in the two deposited films have a low probability of aligning up with each other (Fig. 25), thereby reducing the leakage level and increasing the TDDB lifetime.

8. Summary and recommendations for further study

A comprehensive insight to the breakdown phenomenon in high- κ dielectric films has been provided in this study. We have used electrical, physical and statistical tools and techniques to investigate the five different regimes of degradation/breakdown ranging from the SILC $\rightarrow$ TDDB (SBD) $\rightarrow$ Di-BD $\rightarrow$ An-BD $\rightarrow$ HBD (thermal runaway). The mechanism underlying the degradation trends in each of these regimes was explored. Physical analysis using HRTEM and STM/CAFM came in very handy in proving that oxygen vacancies are the fundamental precursors for high- κ stack failure and the grain boundary plays a critical role in the kinetics and stochastics of the failure distribution. Table 2 provides a gist of the five different breakdown regimes and the mechanisms and specific features representing each of them. Though these five regimes theoretically exist, only the SILC, TDDB (SBD) and Di-BD

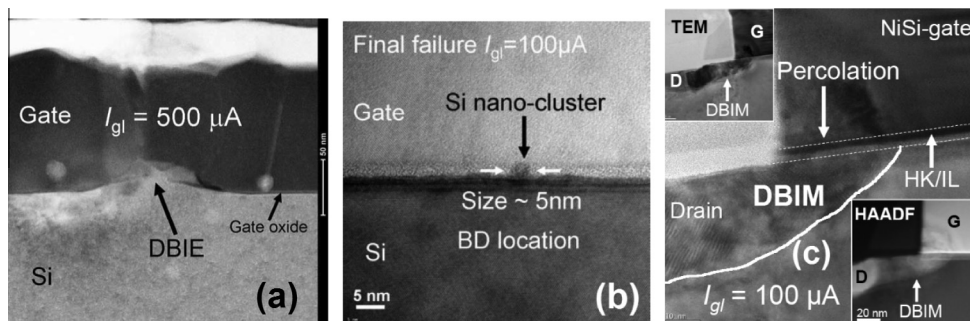


Fig. 22. Defect morphologies for very severe HBD where (a) DBIE protrudes all the way from substrate to gate, (b) Si nanowire like structure is formed at the percolation point for $SiO_2/SiON$ based device [53] and (c) lateral diffusion of nickel from drain contact into the channel region (DBIM) [57]. A similar diffusion from the source end can cause a complete channel short.

Table 1

Failure defects observed in various gate material stacks for Soft and Hard Breakdown and their driving forces.

Gate material	Failure defect in oxide	Driving forces
All gate stacks (SBD)	Si–O/Hf–O bond breaking Generation of oxygen vacancies Percolation path formation Grain boundaries in high- κ are weakest link	Electric field Applied temperature
Polysilicon gate – SiO ₂ /high- κ (HBD)	DBIE and silicon hillock formation Lateral dilation of percolation path resulting in silicon or hafnium nanocluster (fully oxygen depleted percolation core)	Current density Local temperature Electron wind force Thermomigration
Nickel silicide gate – SiO ₂ /high- κ (HBD)	Metal filamentation (Nickel spiking in the [111] direction all the way into the silicon substrate) DBIM – lateral diffusion of Ni from the silicide contacts – this can lead to ‘channel short’ in the extreme case Breakdown is partially reversible by joule heating assisted filament rupture; however, transistor functionality may not be restored	Ion/hole migration Temperature gradient Concentration gradient
TaN/TiN – SiO ₂ /high- κ (HBD)	Metal Filamentation (Tantalum forms bowl-shaped large filaments in the dielectric. Isotropic diffusion of Ta) Irreversible breakdown as the filament size is large	Ion/hole migration Temperature gradient Concentration gradient

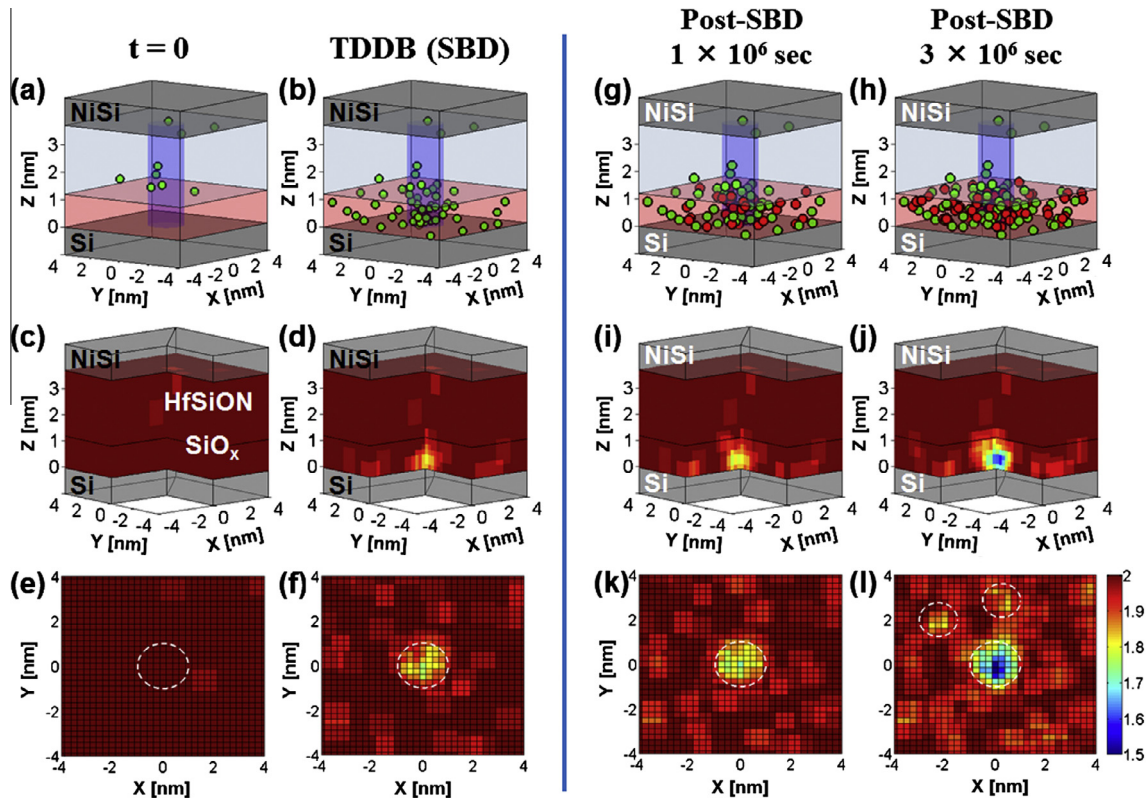


Fig. 23. Simulation of trap generation and conduction at $V_{op} = 1$ V in a HK (25 Å) – IL (12 Å) dual layer dielectric stack, where the green and red defects refers to pre-TDDDB and post-TDDDB traps respectively. The simulations clearly show that the IL breaks down first beneath the HK GB and even subsequent to first BD event, the next BD events also occur in the same IL layer, while HK remains robust. The number of defects generated in the HK layer is negligibly small [61]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

stages are of practical relevance when considering a circuit operating at $V_g = 1$ V. Our analysis also showed that circuit level failure only occurs by multiple SBD events in the IL layer and that the IL layer is the first to break down (it has a lower bond activation

energy (~ 2.3 eV) [71] than HK (4.4 eV for HfO₂) [60] and higher ξ -field distribution).

There is still scope for further work on high- κ breakdown. One area of interest that deserves more focus is the change in

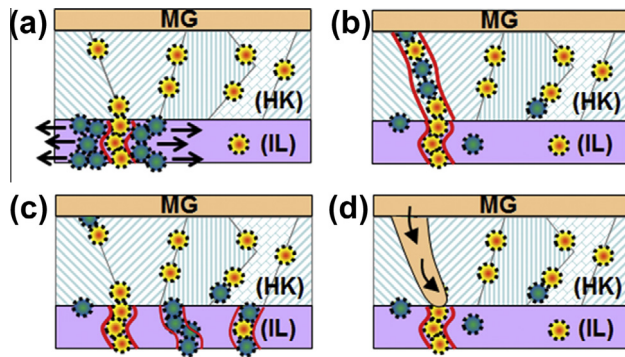


Fig. 24. Four possible scenarios of breakdown in MG-HK stacks. This include (a) lateral dilation of the percolation region confined to the IL layer, (b) full percolation across the HK-IL stack, (c) multiple SBD events within the IL layer and (d) metal filamentation [61]. Scenario C is the most likely event, followed by Scenario A. However, Scenarios B and D are not at all feasible at $V_{op} = 1$ V. The yellow and green circles represent pre-SBD and post-SBD traps generated respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mechanism (if any) when the Si substrate is replaced by III-V materials such as GaAs, InP for high speed (mobility) – high power applications. Since most III-V materials do not have a native oxide,

the interface between the HK dielectric and substrate has very high defect density. In such cases, the kinetics of degradation may be quite different from that of silicon. Although we only studied the TDDB phenomenon here, other failure mechanisms such as negative/positive bias temperature instability (NBTI/PBTI) [72] and hot carrier injection (HCI) [73] also degrade the transistor performance. Some of these mechanisms have a common driving force and they may even be dependent on each other. Stochastic modeling of the interaction of these front-end failure mechanisms and the role of their dependency on further lifetime reduction is very much required. A robust and generic reliability model framework that collectively accounts for all these failure mechanisms and their relative criticality would be useful. While we use the thermochemical model most frequently to talk about defect generation in the high- κ , it fails to explain the polarity dependence of breakdown [74]. As a result, charge-fluence based models should be combined with field-based thermochemical model to be able to better explain the observed polarity dependence. Studies probing the trend of breakdown in FinFETs, GAA-FETs and other novel architectures is still in its incipient stages and this requires attention as well.

It is necessary to propose different DFR initiatives that can help to extend the lifetime of ultra-thin high- κ films by as much as possible. One of the DFR options we have studied is the selection of

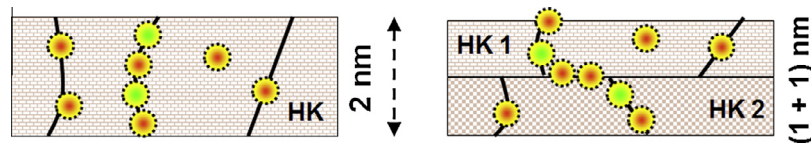


Fig. 25. Use of multi-stage high- κ film deposition as a process design for reliability (DFR) tool so that the grain boundaries are misaligned with each other and number of defects needed for percolation is increased. In the above illustration, while 4 defects could percolate the stack on the left, we need 6 defects for the breakdown of the dual layer stack on the right, although both films have the same total physical thickness. The GB misalignment helps enhance TDDB robustness here.

Table 2

Summary of the different regimes of breakdown and their mechanism of degradation.

Regime	Mechanism of degradation	Remarks
SILC	Oxygen vacancy (defect) generation across the whole dielectric Defect clustering during later stages as more traps nucleate	Area dependent Presence of grain boundary can localize the defect generation process Higher trap generation rate in the IL than the HK Transistor performance acceptable
	Preferential defect generation within the interfacial layer	
TDDB (SBD)	Percolation event due to chain of vacancies aligned in a certain fashion In most cases, IL is the first to fail for a dual-layer HK-IL stack	Area dependent and Weibull distributed for amorphous dielectric Deviation from Weibull trend and area scaling expected for polycrystalline dielectric Transistor performance acceptable
Di-BD	Electron capture and emission at trap sites in the percolation path	Area independent, local phenomenon Mean current remains constant. If N discrete levels of current are observed, we have $\log_2(N)$ number of active defects Transistor performance acceptable, but variability due to RTN may be high
An-BD	Positive feedback between current and temperature and additional thermochemical defect generation around percolation path causes it to dilate The high local current density can also induce DBIE defect due to silicon hillock formation, leading to oxide thinning	Area independent, local phenomenon driven by Joule Heating Monotonic increase in current, low noise Transistor performance severely degraded and unacceptable
HBD	DBIE and Si or Hf-based nanoclusters can form in the percolation core For MG-HK, filamentation will occur. DBIM can also exist due to lateral diffusion of silicide contact material	Ion (Hole) migration Diffusion/drift of ionic species Chemical potential/temperature gradient is a key driving force Transistor is “dead”

metal gate electrodes with high oxygen solubility (serving as good oxygen reservoirs) so that the SILC and TDDDB damage can be partially and repeatedly recovered. The second DFR option is the multi-stage deposition of HK film so that the GB fault lines are purposely misaligned. The last option is to always have a HK–IL dual layer stack without pushing for zero-IL devices because the HK acts as a good voltage buffer [61] and dampens the wear-out kinetics of the percolation path within the IL.

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